Sector-Level News Sentiment and Short-Horizon Return Sensitivity: Evidence from U.S. Technology Stocks

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0 Abstract

This thesis examines whether sector-level news sentiment predicts next-day excess returns for U.S. technology stocks and whether short-horizon sensitivity varies systematically across firms. Using RavenPack's Event Sentiment Score (ESS), I construct daily positive, negative, and net sentiment indices for the technology sector, along with leave-one-out (LOO) versions that exclude each firm's own news when measuring its exposure to industry-wide tone. The empirical analysis is based on a firm-day panel of 681 U.S.-listed technology firms from 2021--2024, merged with CRSP returns and Fama--French factor data. I estimate predictive regressions of day $t+1$ excess returns on sector sentiment, controlling for standard risk factors and firm characteristics, with firm fixed effects and two-way clustered standard errors by firm and date.

Baseline results show limited average next-day predictability from sector-level sentiment, whether measured via aggregate or LOO indices or decomposed into positive and negative components. This pattern is robust to alternative inference procedures, factor controls, subperiod splits, weekly aggregation, winsorization, and coverage-breadth adjustments. Under the baseline ESS construction, heterogeneity tests indicate that smaller firms exhibit stronger sensitivity to sector sentiment and that the marginal effect declines with firm size; however, the strength of this size-related heterogeneity becomes weaker when sentiment is re-measured using headline-based

classifiers. Overall, the evidence suggests that in an information-dense, expansionary environment, sector-wide news is incorporated quickly on average, while smaller technology firms can be more exposed to transient sentiment-driven price pressure under richer, event-level sentiment measurement.

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1 Introduction

1.1 Research motivation and background

Daily price movements in U.S. technology stocks often look more closely tied to shifts in market mood than to the slow-moving fundamentals that ultimately determine cash flows. This impression is not new, but it has become more salient in the post-pandemic period. In 2021, reopening optimism and abundant liquidity helped fuel broad technology rallies; in 2022, fear around inflation, rate hikes, and earnings compression produced sharp reversals; and from late 2023 through 2024, a new wave of AI-related enthusiasm again lifted the sector. Over these episodes, media narratives, headlines, and real-time commentary changed quickly—sometimes within a single trading day—while firms' underlying prospects changed far more slowly.

Against this backdrop, it is natural to ask whether industry-level news tone contains short-horizon information for returns, or whether markets absorb it essentially on impact. Put differently: when day-to-day coverage of the technology sector turns more positive or negative, do prices systematically drift the next day, or has the information already been impounded by the close? This is both a practical and a conceptual question. For investors, even small next-day effects matter for risk management and execution. For researchers, short-horizon links between sentiment and returns speak to the speed and completeness of price discovery, investor attention, and limits to arbitrage in a high-information-flow setting.

Technology is a useful laboratory for this question. Compared with many other sectors, technology firms generate a high volume of discrete news: product launches, regulatory updates, security incidents, venture and partnership announcements, and macro-narratives about platforms and ecosystems. These items often receive wide coverage from general financial media and specialized outlets. The sector is also characterized by substantial heterogeneity—large platform companies alongside small, less-covered firms—which may lead to systematic differences in how prices react to common information. Finally, because technology companies are highly visible to retail and institutional investors, sentiment shocks at the industry level can plausibly diffuse quickly across individual names. Whether such diffusion leaves any predictable next-day imprint is an open empirical matter and the central focus of this thesis.

1.2 Concept and empirical setting

This study examines the next-day return sensitivity of U.S. technology stocks to industry-wide news sentiment. I measure daily tone using RavenPack's Event Sentiment Score (ESS), a standardized metric that classifies news items as positive or negative and maps them to firms and sectors. Rather than studying firm-specific headlines, I aggregate to the sector level to capture the broad mood surrounding the industry on a given day.

ESS is used as the baseline operationalization of daily sector-level sentiment; in robustness analyses, I verify that the main conclusions are not driven by a particular sentiment measurement approach by considering alternative sentiment measures.

A design choice is to separate the general sector tone from any self-reaction to a firm's own news. To do this, I construct a leave-one-out (LOO) sector sentiment index: for each firm-day observation, the sector index is recomputed excluding that firm's own mapped items. The LOO construction mitigates two common concerns. First, large firms tend to dominate sector aggregates simply because they generate more coverage; excluding a firm's own items reduces this dominance when estimating that firm's exposure. Second, if a firm releases news late in the day and its price adjusts immediately, the next-day regression could mechanically reflect the tail of that firm's own reaction; removing self-news helps focus on the influence of external, industry-level tone. Throughout, I compare results using the broad sector index and its LOO analogue to assess whether any observed next-day sensitivity is driven by common sentiment rather than by firm-specific spillovers.

The empirical horizon is deliberately short. I relate day t sector sentiment to day t+1 excess returns, aligning news items to U.S. trading sessions and rolling post-close items to the next day's session. Short horizons sharpen the test of immediate absorption versus a one-day drift and limit contamination from unrelated macro or firm events. At the same time, short horizons imply that effect sizes, if present, are likely to be small and sensitive to microstructure frictions, such as liquidity conditions, inventory risk, and attention constraints.

1.3 Literature gap and theoretical expectations

A large literature connects media tone and investor sentiment to short-horizon return dynamics. Tetlock (2007) documents that high pessimism in a major column predicts market rebounds the following day, consistent with temporary mispricing that reverses once attention rebalances. Subsequent work shows that predictability can depend on economic conditions: Garcia (2013) finds stronger effects in recessions, when risk appetite and attention are strained, and weaker effects in expansions. These studies typically analyze broad market proxies or mixed-industry samples and rarely isolate a single sector over a defined recent window.

Several mechanisms suggest why one might expect limited next-day predictability in an expansionary, high-liquidity environment such as 2021–2024 for technology. First, high investor attention and abundant intermediation could speed the incorporation of widely discussed sector news, leaving little room for next-day drift. Second, algorithmic and headline-sensitive trading may exploit sentiment signals within minutes or hours, especially for large, well-covered firms. Third, to the extent that many sector headlines reiterate public information, their marginal informational content may be low. These arguments push toward a null of little average next-day effect.

On the other hand, two countervailing channels point to heterogeneous responses across firms. Limits-to-arbitrage and attention-based models suggest that smaller, less-covered stocks may react more slowly to common information because trading is costlier, inventory capacity is thinner, and investor attention is scarcer. In addition, when sector tone becomes strongly positive or negative, portfolio rebalancing and ETF flows can transmit shocks mechanically across names, with the impact larger where liquidity is thinner. If these mechanisms matter, we may observe stronger next-day sensitivity for small-cap technology stocks, particularly when sentiment measures capture richer, event-level information.

Taken together, prior evidence and theory motivate two expectations that guide the empirical design: (i) on average, industry-level sentiment shocks in the technology sector should be absorbed quickly, implying small or insignificant next-day coefficients; (ii) cross-sectionally, sensitivity may be stronger among smaller firms (and potentially among firms with higher volatility or lower liquidity), with the strength of such heterogeneity depending on how sector-wide sentiment is quantified. The LOO construction helps test whether any such sensitivity reflects exposure to a sector-wide mood rather than a firm's own headlines, though cross-sectional patterns may still be attenuated when sentiment is measured using headline-only classifiers. A further implication is that any conclusion about rapid absorption (or size-dependent frictions) should be robust to reasonable alternative sentiment measurement choices, rather than hinging on a single scoring algorithm.

1.4 Contributions of this study

This thesis provides three contributions, each intentionally modest. First, it offers a focused sector‑level test of short‑horizon sentiment effects using a leave‑one‑out design that separates common tone from firm‑specific news. Most prior work uses market‑wide proxies; a dedicated technology‑sector analysis clarifies whether common sentiment travels similarly within a single, information‑dense industry.

Second, it provides evidence that cross-sectional sentiment sensitivity is concentrated among smaller firms under structured, event-level sentiment measurement (ESS), while such heterogeneity attenuates under headline-based constructions. This highlights that detecting differential firm responses may require richer sentiment signals than headline-only tone measures.

Third, the study illustrates a practical template for integrating sector sentiment into short‑horizon asset‑pricing panels. The implementation combines sector/LOO indices with firm‑demeaned regressions and two‑way clustered standard errors by firm and date. This combination is straightforward to apply in other settings and helps align inference with the panel's structure.

1.5 Research design overview

The empirical analysis uses a daily firm‑level panel from 2021 through 2024 covering 681 U.S. technology stocks. I merge CRSP returns, firm characteristics, Fama–French six factors, and RavenPack news items mapped to technology firms. Each news item is time‑stamped to U.S. trading sessions; items after 16:00 Eastern are rolled to the next trading day. Daily sentiment indices (positive, negative, and net) are constructed at the firm level and then aggregated to sector‑day measures. For each firm‑day, I compute the corresponding LOO indices by excluding that firm's own mapped items from the sector aggregate.

The baseline specification relates next‑day excess returns to sector sentiment and LOO sentiment, controlling for market factors (MKT–RF, SMB, HML, RMW, CMA, and MOM) and standard firm characteristics (size, recent volatility, turnover, and a rolling beta). Coefficients are estimated with firm fixed effects, and standard errors are two‑way clustered by firm and date, following common practice in high‑frequency asset‑pricing panels. To probe mechanisms, I interact sentiment with indicators for size, volatility, beta, and turnover, and then replace the indicators with standardized continuous traits to assess monotonic patterns. Robustness checks examine alternative clustering schemes, winsorization thresholds, subperiod splits, and a weekly aggregation that smooths idiosyncratic noise.

Two design features deserve emphasis. First, by focusing on t to t+1 links, the study asks a narrow question about one‑day drift after industry‑level mood changes. This keeps interpretation clean: small coefficients are consistent with rapid absorption; larger ones indicate slower diffusion or frictions in trading and attention. Second, the LOO approach makes the exposure truly external. It ensures that the sentiment variable capturing "industry mood" for a given firm is not mechanically influenced by that firm's own headlines on the same day.

1.6 Main findings and structure of the thesis

The main results are straightforward. The absence of average next-day predictability is consistent with heterogeneous firm responses to sector-level sentiment. This pattern fits the idea that, in a well‑covered, expansionary period, industry news is incorporated quickly. At the same time, the response to sector sentiment is not uniform across firms. Under the baseline ESS sentiment construction, smaller firms exhibit stronger sensitivity to both positive and negative sector tone, while larger firms show little response. This size-related pattern is evident in both aggregate and leave-one-out (LOO) indices under the ESS-based construction, suggesting that it reflects exposure to common industry sentiment rather than mechanical self-news effects. However, the strength of this heterogeneity is sensitive to how sentiment is measured and becomes weaker under alternative, headline-based sentiment constructions.

These findings have two implications. First, on average, daily sentiment shocks in the technology sector seem to be quickly incorporated, limiting their role as a broad predictor of the next day. Second, heterogeneity is important: even when the average effect is small, under the ESS based structure, some companies, especially small companies, show a response consistent with friction and attention restriction. For practitioners, this shows that they should be cautious when inferring the sentiment signals at the industry level without taking the company characteristics as the condition. For researchers, the results point to the value of design, which allows public information to have an uneven impact among companies.

The remainder of the thesis proceeds as follows. Section 2 reviews related literature on media tone, investor sentiment, and short‑horizon return dynamics, with an emphasis on mechanisms that predict heterogeneity across firms. Section 3 describes the empirical methodology, including the construction of sector and LOO sentiment indices and the panel regression framework with two‑way clustered inference. Section 4 details data sources and variable construction. Section 5 presents the baseline and heterogeneity results, along with robustness checks. Section 6 discusses implications and limitations, and Section 7 concludes.

1.7 Notes on terminology and scope

To keep the discussion focused, I use ‘sentiment’ to refer to the observed tone of mapped news items, not to unobserved investor beliefs. In the baseline analysis, this tone is summarized using RavenPack’s Event Sentiment Score (ESS); robustness checks consider alternative sentiment measures.“Predictability” refers to statistical association between day t sector tone and day t+1 excess returns, not to tradable profitability after costs. The sample is restricted to U.S.-listed technology firms with valid mappings in the data sources; results should be interpreted within this scope.

2. Literature Review

2.1 Investor Sentiment and Market Predictability

For a long time, investor sentiment has been regarded as a short-term driver of asset prices and cannot be fully explained by fundamentals. Baker and Wurgler (2006, 2007) formally established this concept by building an sentiment index based on surveys, IPO activities and capital flows. They found that optimism would push up stock prices that are difficult to arbitrage, such as small, volatile and short-term companies. Brown and Cliff (2004) also pointed out that sentiment is linked with market valuation, but once the price is adjusted, its forecasting ability will be weakened. Schmeling (2009) extended these findings to the international scope, revealing that culture and market maturity determine the intensity of sentiment effects.

However, there are still differences on the persistence of sentiment influence in the literature. Some studies have found that mispricing can be reversed within a few days, while others have observed that sentiment will persist during boom times. This unresolved contradiction - that is, the contradiction between behavioral overreaction and the efficiency of rapid information dissemination - is particularly prominent in the field of information intensive technology, which also leads to the discussion on media channels in the next section.

2.2 News and Media Sentiment in Financial Markets

The media tone is one of the most direct channels for investor sentiment to affect prices. Tetlock (2007) pointed out that the pessimistic language in the financial column indicates a decline on the same day and a partial reversal on the next day, which means that the market has temporarily overreacted. Tetlock, Saar-Tsechansky, and Macskassy (2008) pointed out that the tone of the text in the company news indicates unexpected profits, but it will soon be digested by the price. Engelberg and Parsons (2011) emphasized the division of attention: local investors will trade disproportionately according to the news related to geographical location.

Later studies added clear dimensions of attention and timing. Da, Engelberg and Gao (2011) developed an Internet search agent to measure investors' attention, and found that high attention will amplify the short-term impact of sentiment. DellaVigna and Pollet (2009) showed that the timing of the announcement is very important - the press conference on Friday postponed investors' reaction to Monday. Barber and Odean (2008) pointed out that even without new fundamental factors, "eye-catching" events will promote retail transactions.In contrast, Jeon, McCurdy, and Zhao (2022) demonstrate that a higher volume and changed tone of firm-specific news are significantly associated with increased jump probability and size—effectively showing that material information is incorporated into price jumps rather than gradual drifts. In short, media tone clearly moves prices, and the prevalence of discrete jumps suggests that slow, next-day drift patterns may be increasingly rare in industries with heavy information flow.

In short, media tone will obviously affect prices, but technological progress seems to shorten the response window - which makes it unclear whether one-day price drift still exists in an industry after in-depth analysis.

2.3 Measuring Sentiment: Methods, Validity, and Data Choices

A persistent challenge in such literature is how to reliably measure sentiments. The dictionary based method (Tetlock, 2007; García, 2013) uses fixed dictionaries such as Loughran–McDonald to count positive and negative words. Although these methods are transparent, they often misclassify context specific words commonly used in science and technology news (such as "beat", "crash", "bug"). Newer machine-learning approaches—including FinBERT and other transformer-based models (Mishev et al., 2020; Gomaa, Bennani, & Denecke, 2023)—capture richer linguistic nuance but at the cost of interpretability and replicability.

Commercial analytics such as RavenPack's Event Sentiment Score (ESS) provide standardized, time-stamped scores mapped to firms and industries (Gross-Klussmann & Hautsch, 2011; Smales, 2014). ESS ensures timeliness and consistent entity mapping, making it practical for high-frequency research, though at the cost of algorithmic transparency and large-firm bias. Given my daily, industry-level focus, ESS offers a practical baseline that balances coverage and reproducibility; alternative sentiment measures are informative benchmarks for assessing the robustness of short-horizon findings.

Table 1. Comparison of Sentiment-Measurement Approaches

Approach	Interpretability	Timeliness	Coverage	Reproducibility	Domain Fit
Dictionary (L&M, VADER)	High	Low–Medium	Broad	Excellent	Weak for tech terms
Machine learning (FinBERT)	Medium	Medium	Needs labels	Moderate	Strong but data-heavy
Commercial (ESS)	Low	High	Extensive	Limited	Moderate

The diversity of methods means that sentiment estimation is not interchangeable. Timeliness or differences in text fields may change the empirical conclusions about short-term predictability. Accordingly, this study employs ESS together with a leave-one-out (LOO) adjustment to minimize firm-specific bias and align with the goal of isolating industry-level sentiment shocks.

2.4 Mechanisms and Heterogeneity

Why do sentiment effects vary by company and period? There are mainly two mechanisms: *limits to arbitrage* (Shleifer & Vishny, 1997) and *attention-based diffusion* (Hong & Stein, 1999). When transaction costs are high or venture capital is limited, the time for prices to deviate from fundamentals will be longer. Similarly, when investors process information in order, the diffusion of new signals in the market will also be uneven. Empirical research confirms these predictions - Baker and Wurgler (2006) found that the sentiment exposure of small cap stocks is higher, while Stambaugh, Yu and Yuan (2012) linked abnormal profitability to the sentiment cycle. Da et al. (2011) and Hirshleifer, Lim, and Teoh (2020) further emphasized that information overload itself will delay absorption.

These frameworks collectively mean that with the improvement of liquidity and information efficiency, the sentiment effect should weaken. On the contrary, residual friction should produce predictable heterogeneity. Therefore, this paper examines two expectations:

- H1: on average, the sentiment shock at the industry level is absorbed within the trading day, and the predictability of the next day is low;
- H2: smaller or less liquid technology companies show stronger next day sensitivity, reflecting attention and arbitrage constraints.

This theoretical basis links behavioral finance with high-frequency asset pricing, and urges us to examine heterogeneity, not just overall predictability.

2.5 industry level information and linkage

Most sentiment studies focus on the market or company level, but the industry is often the real center of the common narrative. Barberis, Shleifer, and Wurgler (2005) pointed out that the trend of stocks in the same industry exceeded the fundamentals; Cohen and Frazzini (2008) show that information spreads along the supply chain. Han, Huang, and Shi (2022) shows that the capital flow of ETFs can mechanically transmit the sentiment impact between constituent companies, which further strengthens the view that the industry operates as an information ecosystem rather than an isolated entity. These findings emphasize that the industry is a meaningful unit, and relevant concerns and transactions will produce collective dynamics.

However, measuring industry sentiment will introduce a well-known endogenous problem: the news of a large company may dominate the index and mechanically link its own return with the overall return. Many previous studies have ignored this issue, and secretly confused the unique and common components of the company. The leave-one-out (LOO) design adopted in this paper directly solves this deviation by excluding each company's own news when calculating the industry tone. Conceptually, this method can isolate the external sentiment shock faced by the company on a specific date. This is consistent with the exogenous information channel logic proposed by Cohen and Frazzini (2008), and makes it suitable for the modern text analysis environment.

By viewing the industry as an "information ecosystem" rather than a simple classification, this analysis tests whether collective sentiment will spread among peers and whether this diffusion will leave a one-day mark that traditional market level research may ignore.

2.6 Summary and Research Gap

In general, previous studies have confirmed that sentiment will drive the market, but there are differences on its persistence and cross-sectional influence range. The progress of information technology has greatly shortened the response time, making it unclear whether the fluctuation of the next day still exists in the widely studied industry. Existing studies have also confused the sentiment of specific companies and the entire industry, limiting the inference of communication between related companies.

This paper makes up for these deficiencies by combining the standardized daily sentiment data with the design of leave-one-out (LOO) method to isolate external industry sentiments. It further examines how sensitivity changes with company size and liquidity - two key indicators of attention and arbitrage friction. From the perspective of behavior and microstructure, this study aims to clarify the speed and imbalance of information absorption of modern technology stocks.

3 Methodology

3.1 Research Strategy

Our research strategy is motivated by the question of whether aggregate news sentiment exerts short-term effects on technology stock returns, and whether these effects differ systematically across firms. To answer this, I adopt a panel-data asset pricing framework that combines sector-wide sentiment shocks with firm-level return sensitivity.

First, I use news analytics data (RavenPack) to construct sector-level sentiment indices. Focusing on sector-level rather than firm-specific sentiment allows us to capture industry-wide information shocks rather than obvious firm-news reactions (e.g., Apple’s bad news lowers Apple’s stock price). This design isolates the spillover effect of aggregate sentiment that is not tied to a single company. To mitigate the problem that a few large firms dominate sector averages, I further construct leave-one-out (LOO) measures, which exclude each firm's own news when computing its exposure.

Second, I adopt a predictive regression approach, where next-day excess stock returns are regressed on sentiment indices, controlling for risk factors (Fama–French six factors) and firm characteristics (size, volatility, beta, turnover). This strategy allows us to test whether news sentiment provides incremental predictive power for returns, beyond what is explained by standard asset-pricing factors (here implemented as the Fama–French six-factor model; alternative factor sets such as FF5 are examined for robustness in Section 5.4.4) and fundamentals.

This study adopts a unidirectional predictive panel framework that relates sector-level sentiment at day t to firm-level excess returns at day t+1. This design aligns with the research objective of assessing short-horizon return sensitivity to industry-wide news, rather than modeling feedback between returns and media tone, as in market-level VAR approaches such as Tetlock (2007). The leave-one-out (LOO) construction further mitigates mechanical links between a firm's own returns and its measured sentiment exposure. Consistent with this design, my aim is not to establish causal effects but to assess conditional associations—whether sector-level sentiment is statistically related to next-day excess returns after controlling for standard risk factors and firm characteristics.

Third, I recognize that the effect of sentiment may not be homogeneous across firms. Prior literature (*Baker & Wurgler, 2006; Stambaugh, Yu, & Yuan, 2012)* shows that smaller, riskier, and less liquid stocks tend to be more sensitive to sentiment. Accordingly, I incorporate heterogeneity tests by interacting sentiment indices with firm characteristics. This enables us to examine whether aggregate sentiment shocks disproportionately affect certain subsets of technology firms.

Overall, this stepwise strategy:

(i) construction of aggregate and LOO sentiment indices

(ii) predictive regressions controlling for risk factors

(iii) heterogeneity analysis—together provide a transparent and rigorous framework to assess whether, and through which channels, industry-wide sentiment shocks are associated with technology stock returns in the short term.

3.2 Model Specification

The baseline regression is:

$$y_{i,t} = \alpha_i + \beta \cdot Sent_t + \gamma' Z_{i,t} + \delta' F_t + \varepsilon_{i,t}$$

- $y_{i,t}$ or $r_{i,t+1} - RF_t$: next-day excess return.
- α_i: firm fixed effects, controlling for time-invariant firm characteristics.
- $Sent_t$: sentiment indices (sector-level or LOO, positive, negative, or net).
- $Z_{i,t}$: firm controls (size, turnover, beta, volatility).
- F_t : Fama–French six factors.
    - **MKT-RF**: market excess return.
    - **SMB**: size factor (small minus big).

- **HML**: value factor (high minus low book-to-market).
- **RMW**: profitability factor (robust minus weak).
- **CMA**: investment factor (conservative minus aggressive).
- **MOM**: momentum factor.

3.3 Estimation Strategy

We estimate the model using firm-demeaned OLS with two-way clustered standard errors (firm and date), following *Petersen (2009)*. Two-way clustering is essential in panel asset-pricing settings because residuals may exhibit both:

- **serial correlation within firms** (due to persistent shocks or overlapping windows in rolling variables)
- **cross-sectional correlation across firms on the same date** (due to market-wide shocks).

Standard errors clustered by firm alone would understate cross-sectional dependence, while clustering by date alone would ignore persistence within firms. *Petersen (2009)* demonstrates that two-way clustering provides consistent inference when both dimensions of correlation are present.

Note: We do not include time fixed effects, since they would be perfectly collinear with aggregate sentiment indices, making it impossible to identify the effect of sentiment.

3.4 Heterogeneity

The effects of aggregate sentiment may not be homogeneous across firms. Prior studies suggest that smaller, riskier, and harder-to-arbitrage stocks tend to be more sensitive to sentiment shocks *(Baker & Wurgler, 2006; Stambaugh, Yu, & Yuan, 2012)*.

To examine whether this pattern holds within the technology sector, I test for cross-sectional heterogeneity in firm responses using two complementary approaches.

First, I implement median-split dummy interactions. On each trading day, the company is divided into two groups according to market value, volatility, beta value and cross-sectional median of turnover. Then, I interact the sentiment index with these dummy variables to test whether the influence of sentiment is different between the two subgroups. (e.g., small vs. large, high vs. low volatility)

Second, the analysis is extended by estimating the continuous interaction model. After determining which firm characteristics showed significant inter group differences in the median-split tests, I further interacted the sentiment index with the standardized continuous measurement of these characteristics.

Each characteristic will be demeaned and divided by its cross-sectional standard deviation before interaction, so that we can capture the monotonous change of sentiment sensitivity along the distribution of firm attributes. This specification refines the group based results by testing whether the heterogeneity observed during continuous simulation of enterprise characteristics continues.

In short, these complementary designs enable us to assess whether the overall sentiment shock will have a disproportionate impact on some technology companies, and whether this sensitivity will systematically change with the key dimensions of the company.

4 Data and Variable

4.1 Data

4.1.1 Data source overview

The sample covers U.S. listed technology firms between January 2021 and December 2024. The choice of this period reflects two considerations:

(i) the post-pandemic market regime, where volatility and sentiment were elevated *(García, 2013*)

(ii) the emergence of the AI boom after 2022, which intensified news coverage and investor attention on technology firms.

We construct a firm-day panel dataset using three data sources:

1. **CRSP daily stock file (dsf):** provides adjusted daily returns, prices, trading volume, and shares outstanding. Firm identifiers are PERMNOs. The sample includes 681 technology firms (based on GICS classification), yielding approximately 738763 firm-day observations (*Fama & French, 1993*).
2. **RavenPack News Analytics:** provides event-level sentiment scores at the entity level, mapped to firms via CUSIP–PERMNO links (*Tetlock, 2007; Sprenger, Tumasjan, Sandner, & Welpe, 2014*).
3. **Fama–French factor library:** daily factor returns (MKT-RF, SMB, HML, RMW, CMA, MOM) and the risk-free rate (RF) (*Fama & French, 2015*).

4.1.2 Why RavenPack as news sentiment data source

This study uses RavenPack's Event Sentiment Score (ESS) as the primary measure of news sentiment. ESS provides structured, time-stamped sentiment at the event level and aggregates consistently across firms and trading days. The measure has been validated in prior market-microstructure and asset-pricing research (Gross-Klussmann & Hautsch, 2011; Smales, 2014; Brogaard et al., 2021). Compared with many transformer-based or lexicon approaches, RavenPack offers a practical combination of transparency, domain validation, and reproducibility—advantages particularly important for academic research. For the theoretical background and methodological comparison, see Section 2.3.

Accordingly, ESS is adopted as the baseline sentiment measure in this study. Importantly, the empirical conclusions are not intended to hinge on the exclusive properties of ESS, and robustness analyses examine whether the main findings persist under alternative sentiment constructions.

4.2 Variables

Dependent variable: The dependent variable is the next-day excess return, defined as

$$y_{i,t} = r_{i,t+1} - RF_t$$

where $r_{i,t+1}$ denotes the daily stock return and RF_t the corresponding risk-free rate.

Firm characteristics (constructed from CRSP):

- ***Size*:** market capitalization = |PRC| × SHROUT (in millions), log-transformed.
- ***Turnover***: daily trading volume / shares outstanding.

- ***Beta***: estimated from 252-day rolling market-model regression of firm excess returns on MKT-RF.
- ***Volatility***: rolling standard deviation of daily returns over the past 60 trading days.

News Sentiment measures (constructed from RavenPack):

We construct daily sentiment indices from RavenPack's Event Sentiment Score (ESS) following three steps:

Step 1 (Firm-level indices): For each firm i and trading day t, I aggregate all preprocessed ESS observations into three indices:

$$POS_{i,t} = mean(max(ESS, 0));\ NEG_{i,t} = mean(min(ESS, 0));\ NET_{i,t} = mean(ESS)$$

This decomposition preserves the sign of sentiment shocks and allows testing for asymmetric effects of positive versus negative news.

Step 2 (Sector-level indices): For each day t, I take the equal-weighted average across all technology firms:

$$POS_t^{sector} = \frac{1}{N_t}\sum_i POS_{i,t}\ ;\ NEG_t^{sector} = \frac{1}{N_t}\sum_i NEG_{i,t}\ ;\ NET_t^{sector} = \frac{1}{N_t}\sum_i NET_{i,t}$$

Sector indices capture broad news sentiment conditions within the technology industry.

Step 3 (Leave-One-Out indices): To avoid a firm's own news dominating its exposure, I construct leave-one-out (LOO) measures by excluding firm i from the sector average:

$$POS_t^{(-i)} = \frac{\sum_j POS_{j,t} - POS_{i,t}}{N_t - 1}\ ;\ NEG_t^{(-i)} = \frac{\sum_j NEG_{j,t} - NEG_{i,t}}{N_t - 1}\ ;\ NET_t^{(-i)} = \frac{\sum_j NET_{j,t} - NET_{i,t}}{N_t - 1}$$

Including a firm's own news in the sector aggregate can mechanically link the sentiment measure to firm-specific return shocks, such as earnings announcements or product releases. To ensure that sector sentiment reflects external industry information rather than a firm's self-news, we compute a leave-one-out (LOO) version when measuring firm i's exposure, excluding firm i from the sector average on day t.

This adjustment reduces endogeneity and mitigates large-firm dominance, preserving meaningful cross-firm variation in exposure to industry-wide sentiment shocks. These measures capture firm-level sensitivity to common sentiment while avoiding the problem that large or frequently mentioned firms disproportionately drive aggregate sentiment (Tetlock, 2007; Sprenger et al., 2014).

4.3 Data preprocessing

Before constructing the sentiment measures, I first identify and match firms across databases to build the final sample.

We begin by identifying firms in the Information Technology sector (GICS=45) from Compustat, yielding 906 distinct GVKEYs. These firms are then linked to CRSP via the CRSP–Compustat link table, which maps Compustat identifiers (GVKEYs) to CRSP securities (PERMNOs). Over the 2019–2024 period, the 906 firms correspond to 914 unique securities (PERMNOs) in CRSP. The discrepancy arises because some firms are associated with multiple securities due to CUSIP changes, corporate restructurings, or multiple listings.

After merging and filtering, I retain 681 active technology securities with valid price and news coverage data for the 2021–2024 window. These securities form the universe for the firm-day panel used in the empirical analysis.

- **Firm Matching**: We link RavenPack news items to firms in the CRSP sample via CUSIP–PERMNO mapping, retaining only events associated with the 681 technology firms in my universe.
- **Event Filtering**: We keep U.S. news events with valid sentiment scores and event relevance ≥ 50 (RavenPack's relevance threshold), discarding low-relevance or missing-score entries. Duplicate (story, entity) pairs are dropped.
- **Time Alignment**: Each news item is assigned to the corresponding trading session. Events reported after 16:00 ET are rolled forward to the next trading day to align with market close.

After preprocessing, daily firm-level sentiment indices are computed and then aggregated into sector and LOO measures, as described in Section 4.2.

This procedure yields firm-day and sector-day sentiment panels that are merged with CRSP returns and firm characteristics.

4.4 Descriptive Statistics

Table 2. Summary statistics of main variables

	count	mean	std	min	1%	5%	25%	median	75%	95%	99%	max
ret_adj	738763.0	0.001	0.053	-0.937	-0.115	-0.059	-0.017	0.0	0.016	0.06	0.135	6.915
sector_net	515089.0	0.051	0.054	-0.069	-0.069	-0.034	0.015	0.047	0.087	0.142	0.196	0.196
loo_net	515089.0	0.051	0.054	-0.123	-0.069	-0.034	0.014	0.047	0.087	0.142	0.196	0.241
size_log	738597.0	20.572	2.553	11.629	15.28	16.386	18.593	20.761	22.335	24.582	26.148	28.996
turnover	738597.0	0.024	0.492	0.0	0.0	0.001	0.003	0.007	0.012	0.045	0.188	118.648
beta_252d	585731.0	1.262	0.643	-3.476	-0.02	0.288	0.88	1.203	1.581	2.385	3.131	9.797
vol_60d	738763.0	0.039	0.034	0.001	0.004	0.014	0.022	0.032	0.047	0.083	0.153	0.91

Table 2 reports summary statistics for the main variables.

Daily excess returns (ret_adj) have a mean close to zero (0.1%), consistent with the notion that expected daily returns are approximately zero under market efficiency. The distribution is tightly centered around zero, with a standard deviation of 5.3%, but exhibits extreme values ranging from –93.7% to +691%. These outliers likely reflect delisting returns or exceptional firm-specific events.

Both sentiment indices (sector_net and loo_net) are normalized around zero, with means of 0.051 and standard deviations of 0.054. Their ranges span roughly –0.12 to +0.24, confirming that sentiment scores capture both strongly negative and strongly positive news events. The similarity between sector and LOO indices suggests that the leave-one-out adjustment does not materially distort the overall sentiment distribution but ensures that large firms do not dominate aggregate indices.

Firm characteristics display substantial cross-sectional variation. Log firm size (size_log) has a mean of 20.6 and spans 11.6–29.0, indicating the presence of both small-cap and mega-cap firms. Turnover is low on average (2.4%), but with extreme values exceeding 100, reflecting occasional episodes of heavy trading activity.

The 252-day rolling beta (beta_252d) has a mean of 1.26, close to the market benchmark, but ranges from –3.48 to +9.80, indicating that some firms display extreme sensitivities to market fluctuations. The 60-day rolling volatility (vol_60d) averages 3.9% per day, but can reach as high as 91% in turbulent periods, consistent with the presence of highly volatile technology stocks.

Overall, the descriptive statistics highlight that while returns and sentiment are centered around zero, firm characteristics exhibit wide dispersion, providing the necessary cross-sectional variation to examine heterogeneity in sentiment effects.

Figure 1: Time series of sector-level net news sentiment and average excess returns, 2021–2024

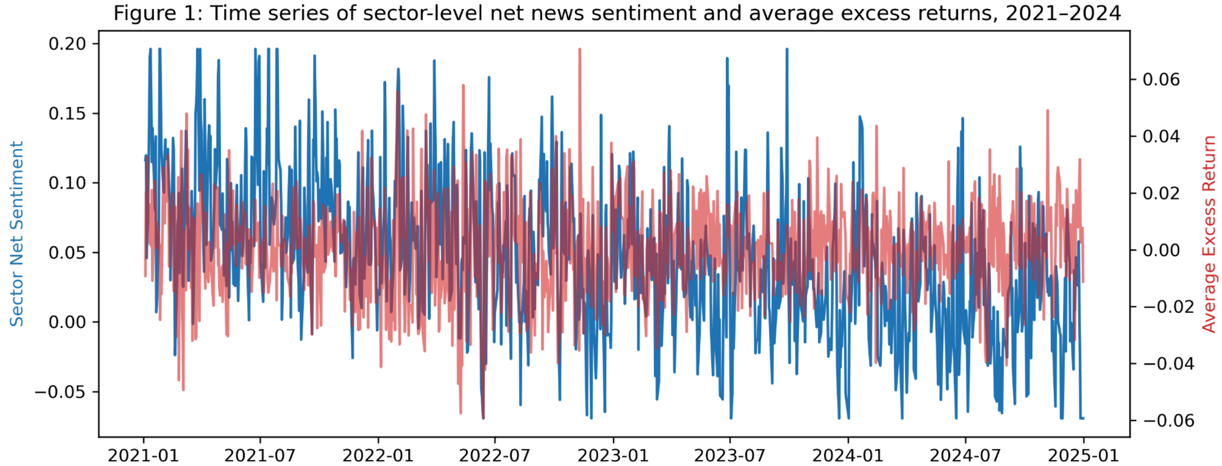


Figure 1 plots the time series of sector-level net news sentiment (blue, left axis) and average daily excess returns of technology firms (red, right axis) from January 2021 to December 2024. Several key features are apparent. First, sector net sentiment exhibits frequent and substantial fluctuations, ranging from approximately –0.05 to +0.20, consistent with episodic bursts of positive or negative news coverage. In contrast, average returns remain tightly centered around zero, rarely exceeding ±5% on a daily basis, reflecting the relatively fast average incorporation of widely disseminated industry-wide news.

Second, although the short-term spikes in sentiment occasionally coincide with market fluctuations–for example, in the case of the tightening cycle in early 2022 and the increase in AI related news in mid-2023– there is no evidence that there is a continuous coordinated movement between sentiment and overall returns. This suggests that, on average, the overall news sentiment shocks are quickly incorporated into the price, and are unlikely to translate into the daily horizon for technology stock returns.

In general, the figure provides a visual preview of my baseline regression results: Although news sentiment has changed significantly, its impact on subsequent market performance seems to be temporary and economically limited.

5 Result

This section presents the empirical results in two parts.

I first document baseline evidence on average next-day predictability from sector-level sentiment.

I then examine whether this average pattern masks systematic cross-sectional heterogeneity in firm responses, and assess the robustness and measurement sensitivity of such heterogeneity.

5.1 Baseline Results

Table 3: Baseline Regressions: Daily Sentiment Measures

Sentiment Type	Coefficient (t-stat)
LOO Positive	0.0104 (0.78)
LOO Negative	-0.0041 (-0.26)
LOO Net	0.0041 (0.46)
Sector Positive	0.0100 (0.74)
Sector Negative	-0.0033 (-0.21)
Sector Net	0.0043 (0.48)

Table 3 summarizes the baseline regressions across sector-level and leave-one-out (LOO) sentiment indices, using positive, negative, and net measures. In all cases, sentiment coefficients are statistically insignificant, with t-statistics close to zero. For example, the coefficient on loo_net is 0.004 (t = 0.46), and the coefficient on sector_net is 0.004 (t = 0.48), neither of which is significant. This baseline pattern indicates limited average predictability and motivates the heterogeneity analysis that follows.

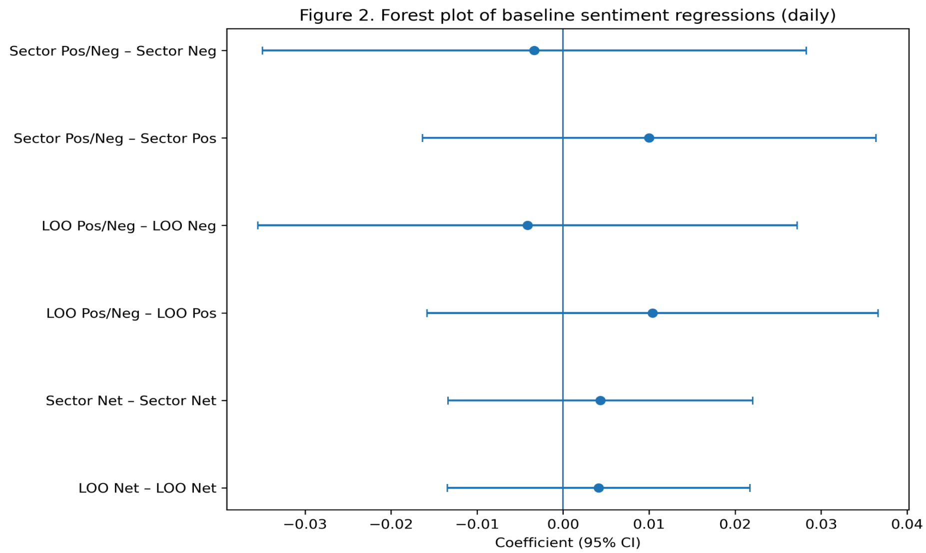


Figure 2 visualizes these results in a forest plot. All confidence intervals cross zero, reinforcing the conclusion from Table 3: aggregate news sentiment—whether measured at the sector level or via LOO indices—does not exert statistically meaningful effects on technology stock returns.

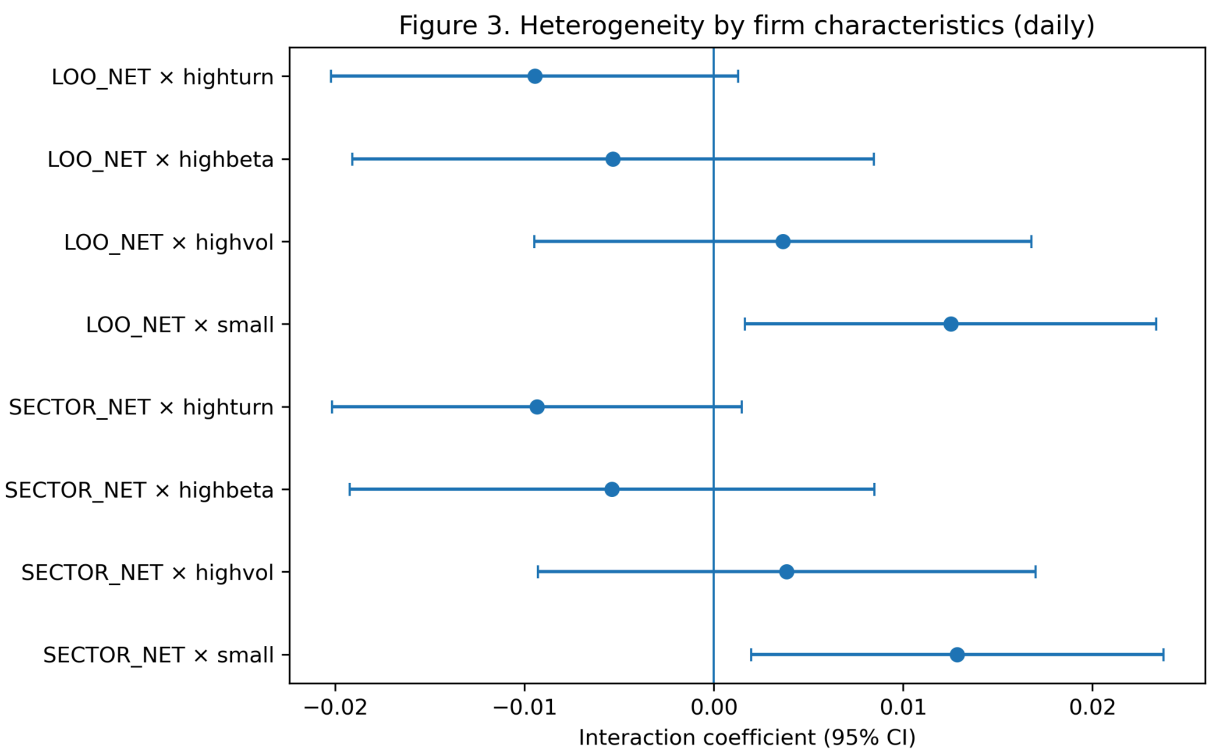
This finding is consistent with the broader sentiment literature highlighting the transient nature of aggregate sentiment effects. *Tetlock (2007)* shows that daily media pessimism predicts contemporaneous downward pressure on market prices, but this effect reverses the following day as prices revert toward fundamentals. Thus, the impact of aggregate media sentiment is confined to the same day, with little predictive power for next-day returns. Our evidence that news sentiment fails to significantly predict next-day excess returns for technology firms is fully in line with this result. *García (2013)* further documents that sentiment effects are more pronounced during recessions but weak in expansions; given that my sample (2021–2024) lies in the post-pandemic recovery and AI-driven expansion rather than a recession, the absence of strong aggregate sentiment effects accords with these prior studies.

5.2 Grouped heterogeneity Result

Table 4: Heterogeneity by Firm Characteristics (Daily)

Specification	Interaction
SECTOR_NET × Small	0.0129 (2.31)**
SECTOR_NET × High Vol	0.0038 (0.57)
SECTOR_NET × High Beta	-0.0054 (-0.76)
SECTOR_NET × High Turn	-0.0094 (-1.69)*
LOO_NET × Small	0.0125 (2.26)**
LOO_NET × High Vol	0.0036 (0.54)
LOO_NET × High Beta	-0.0053 (-0.76)
LOO_NET × High Turn	-0.0095 (-1.73)*

Notes: This table reports interaction effects between net sentiment measures and firm characteristics. Small, High Volatility, High Beta, and High Turnover are indicator variables defined based on sample medians. t-statistics are reported in parentheses. * and ** denote significance at the 10% and 5% levels, respectively.



This subsection examines whether the absence of average predictability masks systematic size-based heterogeneity in firms' responses to sector-level sentiment.

Table 4 and Figure 3 report the grouped heterogeneity results by interacting sentiment with four firm characteristics: size, volatility, beta, and turnover, for both sector-level and leave-one-out (LOO) sentiment indices under the baseline ESS sentiment construction. Firms are divided into two groups each day based on the cross-sectional median of the characteristic (median split).

The interaction Sentiment × Small is positive and statistically significant at the 5% level in both sector and LOO specifications. This finding suggests that small-cap firms are significantly more sensitive to sentiment shocks than large-cap firms, consistent with the literature arguing that sentiment effects are concentrated in "hard-to-value and hard-to-arbitrage" stocks (*Baker & Wurgler, 2006*).

In contrast, the interactions with volatility and beta are small in magnitude and statistically insignificant across both sector and LOO specifications. Turnover interactions are weakly negative and marginally significant at the 10% level, suggesting that higher-turnover firms might respond less to sentiment shocks, though this evidence is not robust. Overall, these results reinforce that firm size is the dominant channel of cross-sectional heterogeneity, while volatility, beta, and turnover do not provide systematic variation in sentiment sensitivity.

5.3 Continuous Interaction Results

To assess whether the size effect documented above reflects a monotonic pattern rather than a binary group difference, I replace the median split with a continuous interaction specification.

In the previous grouped heterogeneity analysis, I employed a cross-sectional median split to divide firms into small-cap and large-cap groups, and found that small-cap firms are significantly more sensitive to sentiment shocks. However, the median split is essentially a binary classification and may lose information on how the effect varies across the continuous distribution of firm size. To further examine whether size heterogeneity changes monotonically with firm size, I standardize firm size within each day's cross-section (size_z_day) and introduce a continuous interaction term, *Sentiment x size_z_day*. This approach not only validates the robustness of the grouped results but also reveals whether sentiment effects gradually attenuate as firm size increases.

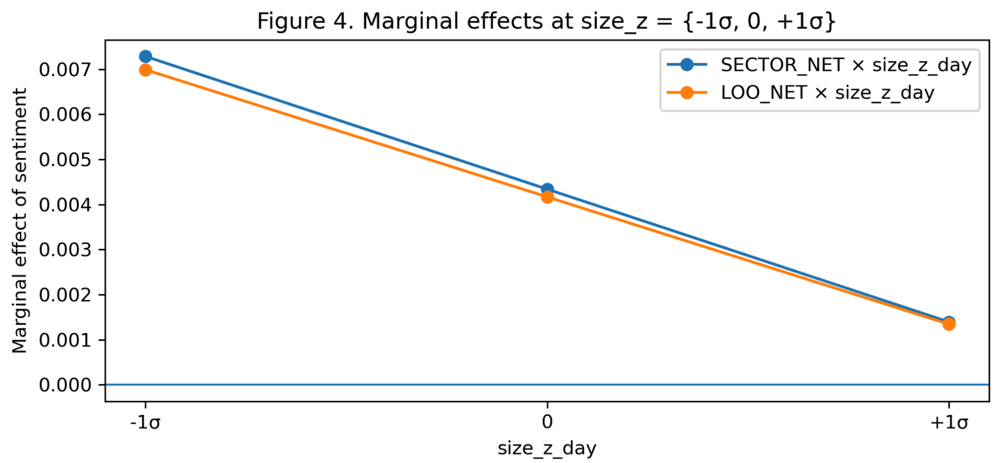
Table 5: Continuous Interaction: Sentiment × Size_z_day (Daily)

Specification	**Sentiment**	**Size_z_day**	**Sentiment × Size_z_day**
SECTOR_NET × Size_z_day	0.0043 (0.48)	0.0001 (0.11)	-0.0029 (-2.46)**
LOO_NET × Size_z_day	0.0042 (0.46)	0.0000 (0.09)	-0.0028 (-2.45)**

Notes: This table reports continuous interaction regressions between net sentiment measures and firm size, standardized at the daily level. All regressions include the full set of baseline controls as well as firm and date fixed effects. t-statistics are reported in parentheses. ** denotes significance at the 5% level.

Table 5 presents the continuous interaction regressions of sentiment with standardized firm size. The interaction terms are negative and statistically significant in both sector-level and LOO specifications (–0.0029, t = –2.46; –0.0028, t = –2.45). These estimates suggest that, under the ESS-based construction, the marginal effect of sentiment declines monotonically with firm size: as firm size increases, sensitivity to sentiment shocks diminishes.

Figure 4 further illustrates the implied marginal effects of sentiment at different size levels (–1σ, 0, +1σ). At one standard deviation below the median (–1σ), the marginal effect of sentiment is about 0.007, while at +1σ it declines to about 0.001. This gradient highlights that smaller firms are much more responsive to sentiment shocks, whereas the effect essentially vanishes for large-cap firms.



Overall, the continuous interaction results reinforce the grouped heterogeneity findings and demonstrate that firm size is the dominant and systematic dimension of heterogeneity in sentiment effects.

5.4 Robustness

The robustness analysis distinguishes between two aspects of the results.
While the absence of average next-day predictability is highly robust across specifications, evidence of cross-sectional heterogeneity is more sensitive to sentiment measurement and therefore examined separately.

5.4.1 Robustness to alternative inference procedures

Our first robustness check evaluates whether the baseline results are sensitive to the choice of standard error clustering. In daily panel regressions, residuals may exhibit both serial correlation within firms and cross-sectional dependence across firms on the same date. To assess the robustness of statistical inference, I re-estimate the baseline specification using three alternative covariance estimators: (i) clustering by firm, (ii) clustering by date, and (iii) Newey–West standard errors.

Table 6: Robustness: LOO — Alternative Inference

	LOO · Firm	**LOO · Date**	**LOO · NW**
LOO_POS	0.0099*** (5.68)	0.0099 (0.76)	0.0099*** (5.39)
LOO_NEG	-0.0025 (-1.08)	-0.0025 (-0.16)	-0.0025 (-1.10)
LOO_NET	0.0041*** (3.20)	0.0041 (0.46)	0.0041*** (3.09)

Notes: This table reports robustness checks for LOO sentiment measures under alternative inference procedures. Columns report standard errors clustered by firm, by date, and Newey–West (NW) standard errors, respectively. t-statistics are reported in parentheses. *** denotes significance at the 1% level.

Table 7: Robustness: SECTOR — Alternative Inference

	SECTOR · Firm	**SECTOR · Date**	**SECTOR · NW**
SECTOR_POS	0.0096*** (5.44)	0.0096 (0.73)	0.0096*** (5.18)
SECTOR_NEG	-0.0017 (-0.73)	-0.0017 (-0.11)	-0.0017 (-0.74)
SECTOR_NET	0.0043*** (3.32)	0.0043 (0.48)	0.0043*** (3.21)

Notes: This table reports robustness checks for sector-level sentiment measures under alternative inference procedures. Columns report standard errors clustered by firm, by date, and Newey–West (NW) standard errors, respectively. t-statistics are reported in parentheses. *** denotes significance at the 1% level.

Table 6 and Table 7 report the results under these alternative clustering schemes. When standard errors are clustered by firm, coefficients on net and positive sentiment (both LOO and sector) are positive and statistically significant. The Newey–West specification produces similar outcomes. In contrast, when clustering by date, none of the sentiment coefficients are statistically significant. Given that the baseline model employs two-way clustering (firm × date), these results indicate that average sentiment–return relations are sensitive to the choice of inference procedure, with limited average predictability under the preferred specification.

Overall, the results are highly sensitive to the choice of clustering. This highlights the importance of using two-way clustered standard errors as the baseline specification, under which sentiment coefficients are statistically insignificant.

Given the firm-day panel structure and the plausibility of both within-firm serial correlation and cross-sectional correlation on the same date, two-way clustering by firm and date is treated as the preferred inference procedure.

5.4.2 Robustness to subperiods

The robustness check examines whether the relationship between sentiment and returns varies over time. To this end, the sample is divided into two periods: before 2023 and after 2023, which roughly captures the beginning of the recent AI-driven market boom. Table 8 presents the estimation results.

Table 8: Robustness: Subperiod Analysis (Pre vs. Post 2023)

	Pre-2023	**Post-2023**
LOO Net Sentiment	0.0053 (0.34)	0.0107 (0.89)
Sector Net Sentiment	0.0055 (0.35)	0.0110 (0.92)

Notes: This table reports subperiod regressions estimated separately for the pre-2023 and post-2023 samples. Each specification includes the full set of baseline controls as well as firm and date fixed effects. t-statistics are reported in parentheses.

For both LOO and sector-level net sentiment, coefficients are small and statistically insignificant in the pre-2023 sample (around 0.005). During the post-2023 period, the coefficients approximately double in size (around 0.010–0.011) but remain statistically insignificant, with t-statistics well below conventional thresholds.

Overall, the absence of statistically meaningful average predictability is preserved across subperiods. The slightly larger coefficients after 2023 may indicate that market sentiment became more closely aligned with short-term return fluctuations during the AI boom, yet these changes are not statistically reliable.

5.4.3 Robustness to weekly aggregation.

To examine robustness at lower frequencies, I constructed an alternative firm–week panel. In this setting, daily excess returns are averaged within each week, while sentiment indices are computed as the mean across trading days of the same week. Firm characteristics are updated using end-of-week values for size and beta, and weekly averages for turnover and volatility. This procedure yields a panel structure comparable to the daily dataset but filters out high-frequency noise.

Table 9: Robustness: Daily vs. Weekly Sentiment Measures

	Daily	**Weekly**
LOO Net	0.0041 (0.46)	0.0019 (0.23)
Sector Net	0.0043 (0.48)	0.0024 (0.29)
LOO Positive	0.0104 (0.78)	0.0060 (0.52)
LOO Negative	-0.0041 (-0.26)	-0.0029 (-0.23)
Sector Positive	0.0100 (0.74)	0.0064 (0.55)
Sector Negative	-0.0033 (-0.21)	-0.0023 (-0.18)

Notes: This table compares regression estimates using sentiment measures constructed at daily versus weekly frequency. Each entry reports the estimated coefficient with the corresponding t-statistic in parentheses. All specifications include the baseline set of controls as well as firm and date fixed effects (defined at the corresponding frequency).

Table 9 presents the results, and Figures C1–C2 in the Appendix provide a visual summary with 95% confidence intervals. At the weekly frequency, coefficients on both LOO and sector-level net sentiment become smaller and remain statistically insignificant. When decomposing sentiment into positive and negative components, the results remain similarly insignificant.

At the weekly frequency, the already weak average relationship between sentiment and returns is further attenuated.

5.4.4 Robustness to control specifications

Table 10: Robustness: Control Sensitivity (FF5 vs. FF6 Baseline)

	FF5 Controls	**FF6 Controls (Baseline)**
Sector Net Sentiment	0.0039 (0.43)	0.0043 (0.48)
LOO Net Sentiment	0.0041 (0.45)	0.0041 (0.46)

Notes: This table compares baseline regression results under alternative factor control specifications. FF5 denotes the Fama–French five-factor model, while FF6 additionally includes the momentum factor. Reported coefficients correspond to net sentiment measures. t-statistics are shown in parentheses.

The momentum factor in the Fama–French six-factor model (FF6) captures short-term return continuation that may overlap with sentiment-driven price movements. To verify that the inclusion of this factor does not absorb the potential sentiment effect, I re-estimate the baseline using the Fama–French five-factor model (FF5), which excludes momentum.

As shown in Table 10, coefficients on both LOO and sector-level net sentiment remain small (≈0.004) and statistically insignificant under either specification. The near-identical results confirm that the baseline finding is not driven by the inclusion of the momentum factor, and that the main conclusions are robust to alternative control sets.

5.4.5 Robustness to winsorization and coverage

Table 11: Robustness: Winsorization and Breadth Adjustment

	Win 1% + Breadth	Win 0.5% + Breadth	Baseline
Sector Net Sentiment	0.0058 (0.64)	0.0058 (0.64)	0.0043 (0.48)
LOO Net Sentiment	0.0055 (0.61)	0.0055 (0.61)	0.0041 (0.46)

Notes: This table reports robustness checks to alternative winsorization thresholds and breadth-adjusted sentiment constructions. Win 1% and Win 0.5% indicate winsorization at the corresponding tails. Breadth adjustment scales sentiment by the cross-sectional coverage of news. Baseline specifications follow the main regression setup. t-statistics are reported in parentheses.

We further test whether the baseline results are affected by outliers or by days with limited firm coverage. Following prior studies such as Tetlock (2007) and Garcia (2013), sentiment indices are winsorized at the 1% and 0.5% tails to mitigate the influence of extreme observations. In addition, I control for the breadth of firm coverage (the number of firms contributing to the daily sentiment index) to account for potential sampling variation in aggregate sentiment.

Table 11 reports the results. Under both winsorization levels, coefficients on LOO and sector net sentiment remain positive, small in magnitude (≈0.005–0.006), and statistically insignificant ($t < 1$). Compared with the baseline specification (≈0.004), the estimates are slightly larger but not meaningfully different in either magnitude or significance.

Consistent with the approaches in *Tetlock (2007)* and *Garcia (2013)*, these results confirm that the absence of significant sentiment effects is **not driven by outliers or sample coverage noise**, but rather reflects the overall lack of a robust return response to daily news sentiment within the technology sector.

5.4.6 Robustness to alternative sentiment constructions

Table 12: Robustness: Alternative Sentiment Measures

	(1) ESS	(2) FinBERT	(3) VADER
LOO Net Sentiment	0.0041	0.0023	0.0019
	(0.46)	(0.32)	(0.29)
Sector Net Sentiment	0.0043	0.0002	0.0035
	(0.48)	(0.04)	(0.29)
Controls	Yes	Yes	Yes
Firm FE (demeaned)	Yes	Yes	Yes
Date FE (clustered)	Yes	Yes	Yes
Observations	515,089	515,089	515,089

Notes: t-statistics in parentheses. All regressions include firm and date fixed effects. Standard errors are clustered at the firm level.

In order to test whether the benchmark regression results depend on the specific news sentiment structure, this paper further uses two alternative sentiment indicators based on news headlines to test the robustness of the benchmark model. Specifically, this paper uses (1) financial text sentiment model finbert based on transformer and (2) sentiment analysis method Vader based on dictionary and rules to replace the proprietary event sentiment indicator (ESS) provided by ravenpack to re estimate the benchmark regression.

For each alternative method, this paper first calculates the sentiment score at the news headline level, and uses the aggregation rules that are completely consistent with the benchmark model to construct the sentiment indicators at the company day and industry day levels. The whole process is consistent with the benchmark regression in terms of sample selection, time alignment, company De mean estimation method and robust standard error of two-way clustering. In addition, this paper also constructs industry level sentiment indicators and Leave-One-Out (LOO) industry sentiment indicators to maintain the identification strategy of distinguishing company specific information and industry common information.

Table 12 reports the benchmark regression results based on Ravenpack ESS, FinBERT and VADER. The results show that under all sentiment measurement methods, the estimated coefficients of industry level sentiment indicators and LOO industry sentiment indicators are small in economic significance, and are not statistically significant. More importantly, the coefficient signs and orders of magnitude under different sentiment construction methods are highly consistent, indicating that the benchmark conclusion is not driven by a specific sentiment model.

In general, given the above results across alternative sentiment constructions, the baseline finding of limited average predictability remains unchanged. No matter the sentiment model based on machine learning or the dictionary method based on rules, the conclusions are consistent, which enhances the credibility of the empirical results of this paper.

These alternative sentiment constructions also provide a natural benchmark for assessing the robustness of cross-sectional heterogeneity, which is examined next.

Table 13: Heterogeneity Robustness: Small-Firm Interaction (ESSvs.FinBERT)

	ESS (Baseline)	FinBERT (LOO)	FinBERT (Sector)
LOO Net × Small	0.0125 (2.26)	-0.0012 (-0.26)	–
Sector Net × Small	0.0129 (2.31)	–	-0.0008 (-0.30)
Controls	Yes	Yes	Yes
Firm FE (demeaned)	Yes	Yes	Yes
Date FE (clustered)	Yes	Yes	Yes
Observations	515,089	515,089	515,089

Notes: t-statistics in parentheses. "Small" is defined using a cross-sectional median split of firm size within each trading day.

I further examine whether the size-based heterogeneity documented in the baseline analysis is robust to alternative sentiment measures. Specifically, I re-estimate the heterogeneity specification using FinBERT-based headline sentiment, maintaining the same regression design, control variables, and fixed-effects structure as in the baseline model.

The interaction term between FinBERT sentiment and the small-firm indicator is statistically insignificant. This suggests that the stronger sensitivity of small-cap firms to news sentiment identified using RavenPack ESS does not robustly carry over to a headline-only sentiment construction.

This finding does not contradict the baseline result. Rather, it highlights an important distinction between sentiment measures. The baseline ESS is constructed from structured, event-level information extracted from full news articles and is designed to capture firm-specific informational shocks. In contrast, FinBERT-based sentiment relies solely on headline text and therefore reflects a more general and noisier measure of news tone.

As a result, cross-sectional differences in sensitivity—such as those related to firm size—may be attenuated when using headline-only sentiment measures. The disappearance of statistically significant heterogeneity under FinBERT thus suggests that the size-based effect documented in the baseline analysis is more closely linked to firm-specific, structured information rather than to generic headline sentiment.

6 Discussion

6.1 Overview of the current study

This paper investigates whether sector-level news sentiment predicts next-day excess returns of U.S. technology firms and whether the effect differs across firm characteristics. Across baseline regressions using both sector and leave-one-out (LOO) sentiment indices, coefficients are small on average, motivating an examination of cross-sectional heterogeneity. However, heterogeneity tests reveal a clear size pattern: smaller firms are more sensitive to sentiment shocks, and the marginal effect declines with firm size. Robustness checks—including alternative inference procedures, subperiod splits, weekly aggregation, alternative control sets, and alternative sentiment measures—confirm the absence of average next-day predictability. The evidence for size-based heterogeneity is strongest under the baseline ESS construction and becomes weaker under alternative sentiment measures.

6.2 Relation to prior literature

The absence of significant next-day return predictability aligns with a growing body of evidence that the market quickly absorbs sentiment-related information. Tetlock (2007) finds that negative media tone induces temporary selling pressure that largely reverses within a day, implying that news sentiment primarily affects intraday trading dynamics rather than next-day pricing. García (2013) further shows that sentiment shocks have stronger predictive power during economic downturns, when investor uncertainty is elevated, but weaken in expansionary periods. Because my 2021–2024 sample largely covers a post-pandemic expansion and an AI-driven market optimism phase, it is unsurprising that I observe little persistence in sentiment effects. The observed size heterogeneity, however, supports Baker and Wurgler's (2006) argument that sentiment plays a greater role for small, hard-to-value, and difficult-to-arbitrage firms. Together, these findings imply that while sector-wide news is swiftly incorporated into prices for large, liquid technology firms, smaller firms remain more exposed to transient sentiment-driven mispricing due to their greater trading frictions and limited investor attention.

6.3 Result Interpretation

Three key patterns emerge from the empirical results.

(1) Baseline results show no evidence of next-day predictability.

Across all baseline regressions using sector and leave-one-out (LOO) sentiment indices, coefficients on net, positive, and negative sentiment remain economically small and statistically insignificant. As reported in Table 3 and illustrated in Figure 2, all confidence intervals overlap zero, implying that technology stock returns do not exhibit delayed responses to sector-level sentiment shocks. This finding is consistent with rapid average price assimilation of widely disseminated sector news: market participants react promptly to aggregate technology news, leaving no systematic drift observable at the daily horizon.

(2) Strong size-driven heterogeneity indicates cross-sectional differences in sentiment sensitivity.

When firm characteristics are introduced through interaction terms, size emerges as the most consistent and economically meaningful channel. In median-split regressions (Table 4), the coefficient on *Sentiment* × *Small* is positive and significant at the 5 percent level, suggesting that smaller technology firms are more reactive to sector-level news. Continuous interaction tests (Table 5, Figure 4) confirm a monotonic gradient: the marginal effect of sentiment decreases steadily with firm size. Quantitatively, a one-standard-deviation decrease in log-size amplifies the sentiment–return slope by roughly 0.007, while large-cap firms display almost zero sensitivity. These results align with the limits-to-arbitrage hypothesis—smaller firms face higher information asymmetry, lower analyst coverage, and greater transaction costs, making them more susceptible to sentiment-driven price pressure (see Figure 4 for implied marginal effects).

This size-based heterogeneity is most pronounced when sentiment is measured using the baseline ESS construction. When sentiment is re-measured using an alternative model-based classifier (FinBERT), the interaction coefficients remain directionally consistent but lose statistical significance, suggesting that the strength of the heterogeneity effect depends on how industry-wide sentiment is quantified.

(3) Temporal and robustness patterns underscore short-lived effects.

Subsample tests (Table 8) show that coefficients are slightly larger in the post-2023 period, coinciding with heightened AI-related media coverage, but they remain statistically insignificant. At the weekly aggregation level (Table 9, Figures C1–C2 in Appendix), all coefficients shrink further in magnitude and lose any residual significance, indicating that sentiment–return relations dissipate when short-term noise is smoothed out. Robustness exercises confirm that these patterns are not artifacts of model specification: using alternative inference methods (firm-clustered, date-clustered, or Newey–West standard errors) or alternative control sets (FF5 vs FF6) does not qualitatively change the outcome. Additionally, the results hold under winsorization of outliers and when controlling for coverage breadth in the construction of sentiment indices. Elevated sector sentiment also tends to coincide with higher turnover (Table 2), suggesting that short-term attention cycles may underlie these transient reactions.

Overall, the evidence suggests limited average next-day predictability from sector-level sentiment in the technology sector, alongside measurement-dependent cross-sectional heterogeneity. At the aggregate level, sector-wide sentiment shocks appear to be incorporated quickly on average, leaving little persistent one-day drift. Under structured, event-level sentiment measures, smaller firms exhibit greater sensitivity to aggregate news flow, consistent with more pronounced but transitory pricing pressure.

6.4 Theoretical and practical implications

Beyond statistical interpretation, these findings hold broader implications for how information is processed and priced in modern financial markets.

First, market structure and information efficiency.

The rapid absorption of sector-level news by technology stocks reflects the high information efficiency of the institutionalized and algorithmic trading market. The lack of strong next day average predictability indicates that for large and well covered companies, any temporary mispricing may be short-term. In this case, news sentiment serves as a real-time information barometer rather than a lasting pricing factor. This strengthens the self correcting nature of competitive and information intensive markets, in which even large fluctuations in media tone cannot produce available excess returns.

Second, heterogeneity and information friction at the firm level.

The significant size gradient highlights how firm visibility and market frictions control the spread of news information. For small and medium-sized companies, limited analyst reports and low transparency make external news reports an important substitute for direct information. Therefore, as a public signal, the sentiment at the sector level is used by investors to infer the fundamentals and amplify the short-term response. In contrast, large cap companies with richer disclosure and stronger liquidity have embedded the most relevant information in the price. Therefore, for opaque companies, media sentiment plays a role of discovery, while for highly visible companies, media sentiment only plays a role of confirmation.

Third, capital-market implications and monitoring value.

Although the overall news sentiment lacks the ability to predict returns, it still captures the collective optimism and attention cycle in the industry. Periods of extreme positive sentiment, such as the artificial intelligence boom in 2023, usually occur at the same time as higher turnover and compressed risk premiums. Therefore, news based sentiment indicators are still valuable for market monitoring: decision makers and portfolio managers can monitor these indicators to determine the active period or excessive attention period before correction.

Overall, these results suggest that although news sentiment is not a pricing risk factor in the traditional sense, it is still a diagnostic tool that can be used to understand how information, visibility and liquidity interact to form the short-term price dynamics of high innovation industries.

6.5 Model Limitations and Identification Issues

This study characterizes conditional associations between sector-level sentiment and next-day excess returns rather than attempting causal identification. Several limitations follow from this design.

First, sentiment and prices may adjust nearly contemporaneously, so even with a t to t+1 alignment, the estimates should be interpreted as predictive correlations rather than structural causal effects.

Second, sentiment measures are subject to classification error and coverage heterogeneity. While the leave-one-out construction mitigates mechanical links and large-firm dominance, cross-sectional patterns—particularly heterogeneity—may be sensitive to how sentiment is quantified.

Third, although the Fama–French factor controls absorb common risk components, they may not span all technology-specific or state-dependent risks. As a result, some sentiment-related estimates could reflect correlated risk or liquidity conditions rather than pure mispricing.

6.6 Limitations and future research

Although this study provides new evidence for how news sentiment at the industry level is related to the returns of technology stocks, there are still some limitations, pointing to potential extensions.

(1) Frequency of Analysis

The analysis was conducted on a daily basis. Although this captures short-term changes, it cannot reveal the intra day assimilation process that takes place in a few minutes or hours. Future work can use higher frequency data, such as intraday prices or event time windows, to determine the instant dissemination of news shocks.

(2) Consistency and Algorithmic Limitations of Sentiment Measurement

Although this study uses RavenPack event Sentiment Score (ESS) as the standard sentiment indicator due to its wide application and effective framework in financial news analysis, different sentiment algorithms may produce inconsistent polarity and intensity evaluation. Lexicon-based methods and Transformer-based models, for example, often diverge when applied to financial texts because of domain-specific vocabulary, context dependence, and differences in linguistic training corpora (Mishev, Kochev, & Gjorgjevikj, 2020). Prior evaluation studies such as *Evaluation of Sentiment Analysis in Finance* (Mishev et al., 2020) demonstrate that even when applied to the same corpus, one model may classify a statement as positive while another labels it as neutral or only weakly positive.

In addition, RavenPack's internal processing pipeline (including entity mapping, event detection, repeated filtering, and correlation or amplitude weighting) may introduce system bias. For example, the algorithm may increase the influence of mainstream media, or underestimate the influence of small companies with limited news coverage (RavenPack Research Team, 2021).

ESS is a relative intensity score, which provides cross company comparability, but may not fully reflect the time evolution or semantic richness of news sentiment.

Although this study uses RavenPack's ESS to achieve consistency and replicability, under the current data structure, it is not feasible to make a complete one-to-one cross model comparison with open text sentiment classifiers (such as FinBERT, VADER, or LLM-based approaches), because RavenPack does not provide access and preprocessing channels for the underlying news text. Therefore, the alternative sentiment measure used in the robustness check is only constructed from the available headline text, while ESS reflects a broader event level aggregation, including additional text and metadata input. Consistent with this difference, under the headline based sentiment measurement, the lack of average predictability on the next day remained unchanged, while the evidence of heterogeneity based on scale became statistically weaker. Future research can incorporate open text sentiment models into comparable data sets to assess the consistency and potential measurement bias of crossover algorithms (Gomaa, Bennani, & Denecke, 2023).

(3) Sector and Period Scope

The sample is limited to the technology industry from 2021 to 2024, which is a period of market optimism. Extending the analysis to other industries or economic downturn can reveal whether the sentiment effect increases when uncertainty increases. Cross country comparison can also test whether the media transparency or investor attention differences will change the speed of information diffusion.

(4) Linearity and State Dependence

Although the model distinguishes between positive and negative sentiment shocks, it assumes that their effects are linear and independent among market states. In reality, the market reaction may be non-linear or state-dependent, extreme negative news depending on the country may trigger a disproportionate response under high volatility conditions. Future research can use threshold, quantile or regime-switching specifications to capture this asymmetric behavior.

(5) Cross-Firm Information Spillovers

The analysis treats firm-level reactions as independent conditional on sector sentiment.. However, in practice, news shocks may spread among companies through the supply chain, technology or investor concerns. The inclusion of this network spillover effect helps to separate the direct sector wide effect from the indirect cross company information transmission. Modeling these links using supply chain data, patent co-innovation networks or shared investor-attention metrics will clarify whether information contagion will amplify or spread the emotional shock at sector level.

In summary, these limitations do not undermine the main conclusions, but highlight the opportunities for further exploration. Combining text analysis, high-frequency data and network-based modeling with a cross sectoral and cross national perspective will further deepen my understanding of how news driven information transmission shapes asset prices in an increasingly digital and interconnected market environment.

6 Conclusion

This thesis set out to answer a narrow but practically and conceptually important question: does sector-level news sentiment about U.S. technology firms predict next-day excess returns, and if so, for which kind of firms? To address this, I combined high-frequency news analytics with a panel asset-pricing framework for 681 technology stocks over 2021–2024. Sector sentiment was measured using RavenPack's Event Sentiment Score (ESS), aggregated to daily positive, negative, and net indices, and complemented by leave-one-out (LOO) versions that strip out each firm's own news when computing its exposure. Within this setting, I estimated predictive regressions of next-day excess returns on sector sentiment, controlling for Fama–French risk factors and key firm characteristics under firm fixed effects and two-way clustered inference.

Three main conclusions emerge from the analysis.

First, Sector-level news sentiment does not generate economically meaningful average next-day predictability across technology stocks. This conclusion also holds when sentiment is re-measured using alternative, headline-based sentiment classifiers. Across specifications that use aggregate or LOO indices, and across positive, negative, and net measures, sentiment coefficients are small and their confidence intervals consistently include zero. This holds under alternative clustering schemes, factor sets, subperiod splits, weekly aggregation, winsorization of sentiment tails, and controls for the breadth of firm coverage. In other words, when viewed at the level of the entire technology sector, news tone appears to be incorporated quickly on average, leaving limited systematic one-day drift in aggregate returns.

Second, the absence of average predictability masks cross-sectional heterogeneity that is most pronounced under the baseline ESS construction, with smaller firms exhibiting stronger sentiment sensitivity.When sentiment is re-measured using alternative, headline-based classifiers, the evidence for size-based heterogeneity becomes statistically weaker, suggesting that this cross-sectional pattern is sensitive to how industry-wide sentiment is quantified. When sentiment is interacted with firm characteristics, size stands out as the dominant dimension along which responses differ. Median-split regressions show that small-cap technology firms are significantly more sensitive to sector sentiment shocks than large-cap firms, and continuous interaction models reveal a monotonic gradient: as firm size increases, the marginal effect of sentiment on next-day returns declines toward zero. For firms one standard deviation below the daily median size, the implied sentiment–return slope is meaningfully larger than for firms one standard deviation above the median. This pattern is consistent with limits-to-arbitrage and attention-based theories, in which smaller, less liquid, and less widely covered firms adjust more slowly and more noisily to common information.

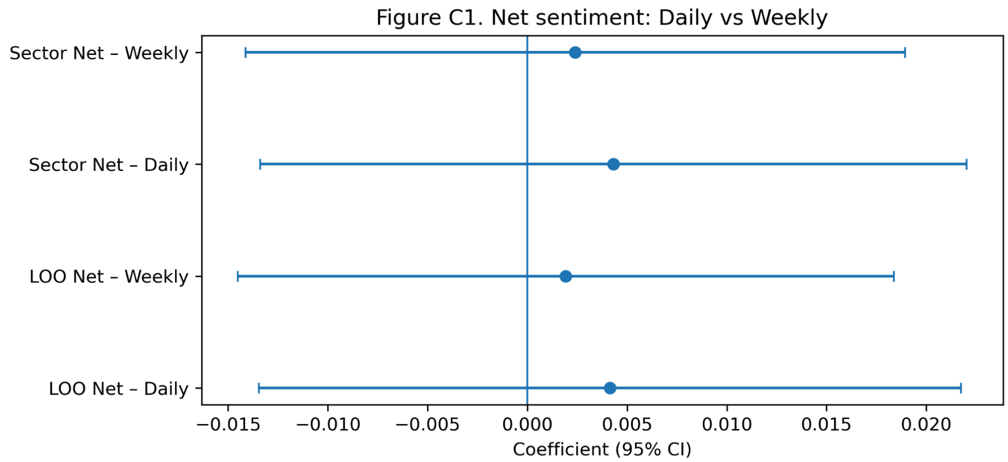
Third, the evidence points to sentiment effects that are short-lived and state-contingent rather than persistent. Subperiod analyses suggest slightly larger (though still insignificant) coefficients during the AI-driven phase after 2023, when technology news volume and investor enthusiasm were elevated, while weekly aggregation further attenuates already small daily effects.Together with the size driven heterogeneity, these results show that the news of the whole industry is more like a catalyst, which promotes short-term rebalancing, especially in smaller companies, rather than as a stable source of predictable returns within a trading day.

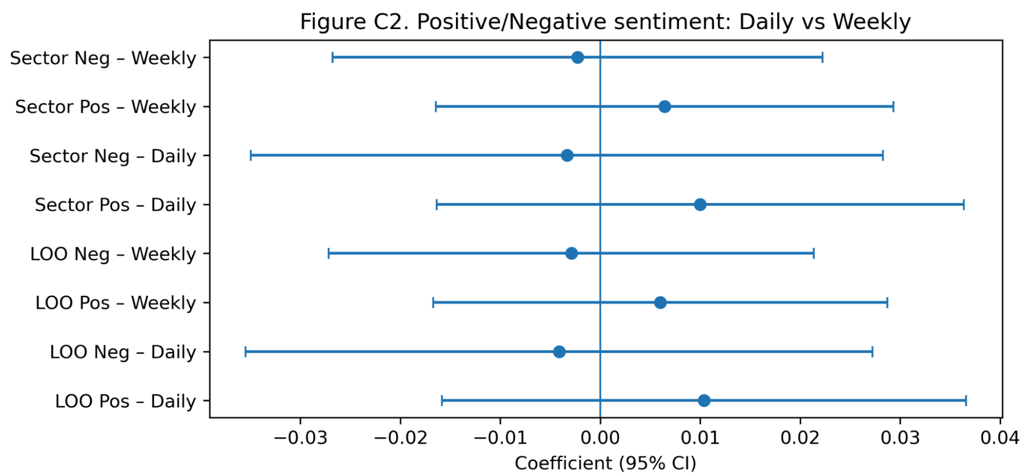
These conclusions have some implications for research and practice. For researchers, these findings emphasize the importance of designs that separates public information from company specific information and allows different firms to react differently. The LOO construction and two-way cluster panel framework used in this paper provides a template for studying the industry sentiment of other industries, markets or time periods. The results also show that in the modern environment with rich information, the aggregate sentiment may not be so important through the slow price drift, but through the intra day dynamics, volatility, and cross-sectional reallocation—dimensions that require higher-frequency data and models explicitly tailored to jump and liquidity dynamics.

For practitioners, the analysis cautions against using a simple industry-level sentiment index as an independent next day trading signal for large and well covered technology companies: once risk factors and friction are taken into account, any available drift is extremely limited. At the same time, the significant scale gradient means that sentiment perception risk management may still be applicable to portfolios with significant exposure to smaller technology stocks, in which news driven flows and temporary mispricing are still more likely. Therefore, monitoring sentiment at the sector level helps to identify periods when such firms are particularly vulnerable to swings in attention and liquidity, even if it does not provide a clear unconditional alpha.

Finally, the limitations of the study - most notably its daily horizon, its dependence on a single proprietary sentiment source, and its focus on one industry and the recent expansion period - leave ample room for future work. Extending the analysis to intraday data, alternative sentiment models applicable to original texts, different industries and more volatile macro systems will deepen the understanding of how news is included in prices. Integrating supply chain links, innovation networks or sharing investor bases can also clarify the channels through which the whole industry sentiment spreads among firms. Among these broader agendas, the evidence provided here provides a key point: in the U.S. technology sector after the pandemic,aggregate news sentiment is priced quickly on average, but its short-term impact is still uneven, and smaller companies bear the brunt of sentiment driven fluctuations.

Appendix





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